

EEG-Based Consumer Preference Prediction Using Logistic Regression

A Binary Classification of Like and Dislike Responses

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ABSTRACT

This study investigates whether electroencephalography (EEG) signals can predict consumer preferences for e-commerce products using logistic regression. Using the publicly available Neuromarketing EEG dataset (1,004 trials from 24 participants), we reframed the problem as binary classification after previous statistical analysis found no significant differences between like and dislike conditions. EEG band-power features from the theta, alpha, and beta bands across 14 electrodes were used to train logistic regression models with subject-wise data splitting to prevent data leakage. Performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. The standard model achieved 56.6% accuracy and a ROC-AUC of 0.602, while the L1-regularized model achieved 57.8% cross-validated accuracy and a ROC-AUC of 0.593. Although both models performed below the majority-class baseline in accuracy, their ROC-AUC values above chance suggest EEG signals contain limited information about consumer preferences. These results indicate that logistic regression captures weak preference-related patterns but is insufficient for reliable prediction, highlighting the need for more informative features or more advanced machine learning models.

CCS CONCEPTS

• Computing Methodologies → Machine Learning

KEYWORDS

EEG Electroencephalography (EEG), logistic regression, machine learning, neuromarketing, consumer preference prediction, binary classification

1 Introduction

1.1 Background and Significance

Consumer preference prediction is a central topic in neuromarketing since it enables researchers and businesses to better understand how consumers evaluate products beyond traditional self-report measures. Conventional approaches rely on conscious responses and may not fully capture the processes underlying consumer decision-making. EEG has attracted considerable attention because it provides a non-invasive and

objective measure of brain activity with high temporal resolution. Previous studies have shown that neural responses recorded during evaluation contain information related to consumer preferences, suggesting that EEG can reveal aspects of consumer evaluation that are not readily observable through conventional behavioral and survey-based methods.

1.2 Theoretical Basis and Motivation

Our previous analysis, conducted in Project I, examined whether frontal alpha asymmetry (FAA), an EEG-derived measure associated with approach-withdrawal motivation, differed between consumers' binary responses. However, no statistically significant difference was observed between the two conditions, suggesting that FAA alone may not be a reliable indicator of consumer preference in this dataset. Rather, consumer preference may be encoded in more complex neural patterns distributed across multiple electrodes and frequency bands, rather than in a single summary measure. This possibility motivates a shift from statistical inference, which focuses on identifying group-level differences, to machine learning approaches that aim to learn predictive patterns from high-dimensional EEG data.

1.3 Research Question and Prediction Objective

This study investigates whether EEG-derived spectral features can predict consumers' product preferences. Specifically, it therefore asks: Can EEG-derived spectral features distinguish between like and dislike responses? To address this question, consumer preference prediction is formulated as a binary classification task using 42 EEG band-power features from the theta, alpha, and beta frequency bands across 14 electrodes. Logistic regression models are trained and evaluated on unseen subjects using subject-wise data partitioning, with performance compared against a majority-class baseline. The results assess the predictive value of conventional EEG spectral features and the limitations of linear machine learning models for consumer preference prediction.

2 Model 1

2.1 Data Acquisition

We used the Neuromarketing EEG Dataset introduced by Yadava et al. This dataset was obtained from the EEG Dataset Repository (github.com/meagmohit/EEG-Datasets) and direct access to the dataset owner was requested to obtain their permission for academic research.

2.2 Method

This test examines whether a 42-dimensional EEG feature set (theta, alpha, and beta power across 14 electrodes) could predict trial-level like/dislike preferences beyond a simple majority-class baseline. Because all the 24 subjects provided multiple trials, we split the data by subject rather than at random: 18 subjects (753 trials) were for the training set and 6 subjects (251 trials) for the test set. Thus, all data of participants only appear in a single data set, with no subject appearing in both sets. This grouped split prevents the model from simply memorizing individual subjects' EEG patterns rather than learning a pattern that generalizes across subjects. Standardization was fit on the training data only and then applied to the test data, to prevent test-set information from influencing preprocessing.

We employed a baseline classifier that always predicts the majority class, which in our case is dislike (accounting for the fact that our data is imbalanced). The accuracy of this baseline classifier is 64.5% (on the test set), which we compare against a logistic regression classifier (scikit-learn, L2-regularized, max_iter = 200, C = 1.0, solver = 'lbfgs'). Logistic regression was chosen because of the ability to interpret the coefficients as changes in log-odds and because of the correlation of the 42 feature sets (34% of feature pairs had $|r| > .8$). Other classifiers like Naive Bayes, which make independence assumptions about the input, would not work as well for this data set. Because the test set was imbalanced, accuracy was not the only performance metric the models were judged on. The models' performances were also measured using precision, recall, F1 score, ROC, and a confusion matrix.

2.3 Results

Descriptive performance statistics are reported in Table 1. The test accuracy (56.6%) of model 1 was below the baseline, and the test ROC-AUC of 0.602 was slightly above the chance level of 0.500.

Table 1. Baseline and Model 1 Performance (N = 251 test trials from 6 held-out subjects).

	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (majority-class)	0.645	—	—	—	—
Model 1 – Train	0.570	0.575	0.304	0.398	0.617
Model 1 – Test	0.566	0.391	0.404	0.398	0.602

The training (57.0%) and test (56.6%) accuracy being relatively close indicates that the model did not overfit on training set samples, but it did not find a strong signal across subjects.

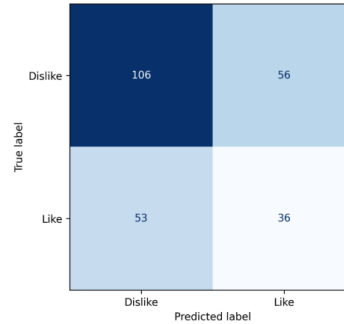


Figure 1. Model 1 Confusion Matrix (test set) N=251 trials from 6 held-out projects

The confusion matrix (Figure 1) indicates that the model predicts 106/162 dislike trials correctly and 36/89 like trials correctly, with the majority class bias expected due to the imbalanced trial type distribution in the test set.

Table 2 lists the ten largest logistic regression coefficients. Theta power at electrodes T7 and T8 showed the strongest positive association with "like," while several beta and alpha features were associated with "dislike." Given that roughly a third of all feature pairs were strongly correlated, these coefficient rankings should be interpreted with caution rather than treated as a definitive ranking of feature importance.

Table 2. Model 1 coefficients (top 10, log-odds).

	Feature	Coefficient (log-odds)	Direction
1	theta_T8	0.751	toward Like
2	theta_T7	0.717	toward Like
3	beta_O2	-0.597	toward Dislike
4	alpha_AF3	-0.597	toward Dislike
5	alpha_AF4	0.474	toward Like
6	theta_F8	0.468	toward Like
7	beta_F4	-0.455	toward Dislike
8	beta_O1	-0.432	toward Dislike
9	beta_F3	-0.426	toward Dislike
10	theta_FC6	-0.408	toward Dislike

Therefore, the near-baseline accuracy, moderate but above-chance ROC-AUC, and the similar training and test accuracies suggest there is little information in the full EEG set for distinguishing like from dislike trials. As noted in Project I, this is consistent with individual differences may dominate over condition-level differences in this sample.

3 Model 2

3.1 Method

Model 1 indicates that the standard logistic regression model using all 42 EEG features exhibited only limited predictive performance. Therefore, we further expanded the analysis by adopting a logistic regression model with L1 regularization and combining it with the nested GroupKFold cross-validation. This model utilized the same 42 frequency band power features extracted during the preprocessing stage, including theta, alpha, and beta power from 14 EEG electrodes. The goal was to predict the "like/dislike" labels for each trial (N = 1,004), while evaluating whether feature selection through L1 regularization could

improve the predictive performance and identifying the most informative EEG features.

As a reference baseline, majority-class prediction (always predicting "Dislike," the more frequent class) yields 56.0% accuracy (563/1,004 trials). We compared this baseline against an L1-regularized ("Lasso") logistic regression classifier, chosen because (a) it performs automatic feature selection by shrinking uninformative coefficients to exactly zero, which is useful given the high feature-to-trial ratio (42 features), and (b) its coefficients remain directly interpretable, allowing us to identify which channel-band combinations contribute most to the prediction.

To prevent data leakage across a given participant's trials and to select the regularization strength C in a way that generalizes to unseen subjects, we used nested, subject-grouped cross-validation. In the outer loop, a 5-fold GroupKFold split partitioned the 24 subjects into five folds (approximately 19–20 subjects per training fold, 4–5 held out for testing), so that no subject's trials appeared in both the training and test sets of the same fold. Within each outer training fold, we used an inner 5-fold GroupKFold to identify the best regularization parameter (C). We evaluated six candidate values (0.001, 0.01, 0.1, 1, 10, and 100) and selected the one with the highest mean ROC-AUC across the inner folds. After selecting the best value of C , we retrained the model using the entire outer training set. For each fold, feature standardization (zero mean, unit variance) was fitted only on the training data and then applied to the corresponding test data. This prevented information from the held-out subjects from influencing preprocessing or model fitting.

Model performance was evaluated using accuracy, precision, recall, F1, and ROC-AUC, averaged across the five outer folds (mean $\pm$ SD), together with a pooled confusion matrix across all folds. We additionally compared training-fold accuracy to test-fold accuracy within the same cross-validation procedure to assess overfitting.

3.2 Results

The L1-regularized logistic regression achieved a mean test accuracy of 57.8% (SD = 2.6%) across the five outer folds, modestly above the 56.0% majority-class baseline. Precision, recall, F1, and ROC-AUC are reported in Table 3.

Table 3. Mean training and test accuracy across the five outer GroupKFold folds

Metric	Value
Train accuracy	0.580 $\pm$ 0.023
Test accuracy	0.578 $\pm$ 0.026
Train–test accuracy gap	0.001

The pooled confusion matrix of all five-fold cross-validation ($N = 1,004$ trials, 24 subjects) is shown in Figure 2. In 441 "like" trials, only 58 were correctly identified (recall rate = 14.1%), while the model correctly identified 522 out of 563 "dislike" trials. This asymmetry indicates that although the model has slightly improved overall

accuracy and AUC compared to the baseline. An ROC-AUC of 0.593, barely above the 0.50 chance level, confirms this weak separation."

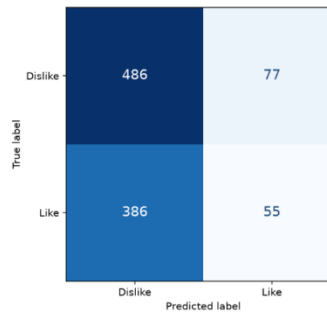


Figure 2. Pooled confusion matrix across 5-fold GroupKFold cross-validation

Examining the mean L1 coefficients across folds (Figure 3), tL1 regularization zeroed 9 out of 42 features, indicating that after applying regularization, the independent predictive information carried by most channel-bandwidth combinations was negligible. Among the retained features, the frontal electrodes (such as FC5, AF4, FC5, F8, etc.) significantly contributed to the top coefficients, which is roughly consistent with the frontal lobe focus measurement examined in the previous section. Though the direction and combination of contributing bands were more heterogeneous than a simple alpha-asymmetry account would predict.

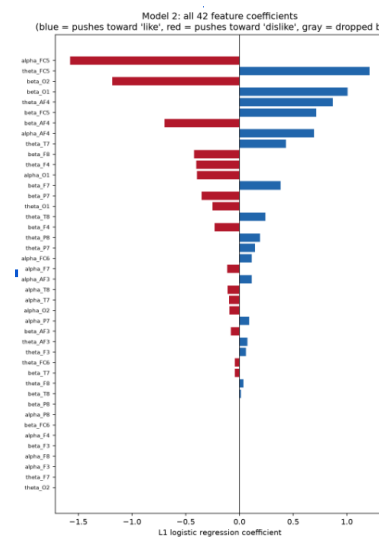


Figure 3. All 42 feature coefficients from the L1-regularized logistic regression, averaged across folds

To assess the overfitting, we employed the same nested cross-validation method and compared the accuracy rates of the training set and the test set. The training accuracy was 58% $\pm$ 0.023, while that of the test set was 57.8% $\pm$ 2.6%. The difference between the two was only 0.1 percentage point, indicating minimal overfitting.

4 Baseline, Model Comparison, and Overfitting Analysis

To evaluate whether the additional feature richness (Model 1) and the L1-regularized, cross-validated approach (Model 2) meaningfully improved on the simplest possible strategy, we compared both models directly against the majority-class baseline under their respective held-out evaluation procedures (Table 5).

Table 4. Baseline, Model 1, and Model 2 performance under their respective held-out evaluation procedures.

Metric	Baseline	Model 1 (test)	Model 2 (5-fold mean $\pm$ SD)
Evaluation setup	—	GroupShuffleSplit, 18 train / 6 test subjects	5-fold GroupKFold, pooled (24 subjects)
Accuracy	0.645* / 0.560†	0.566	0.578 $\pm$ 0.026
Precision (Like)	—	0.391	0.646 $\pm$ 0.228
Recall (Like)	—	0.404	0.141 $\pm$ 0.150
F1 (Like)	—	0.398	0.191 $\pm$ 0.178
ROC-AUC	0.500 (chance)	0.602	0.593 $\pm$ 0.025

*Majority-class accuracy computed on Model 1's held-out test split (162/251 trials). †Majority-class accuracy computed on the full dataset (563/1,004 trials), the evaluation basis used for Model 2. The two baselines differ because Model 1 and Model 2 use different held-out evaluation schemes (a single subject-grouped split vs. pooled 5-fold cross-validation), not because the classification problem itself changed.

Neither model achieves a meaningful improvement over guessing the majority class. Model 1's single-split test accuracy (56.6%) falls below the majority-class baseline computed on that same split (64.5%), while its ROC-AUC of 0.602 remains only modestly above chance. Model 2's five-fold cross-validated accuracy (57.8% $\pm$ 2.6%) exceeds the full-dataset majority-class baseline (56.0%) by only 1.8 percentage points, and its ROC-AUC (0.593 $\pm$ 0.025) is similarly close to chance. Because the two baselines differ, the discrepancy reflects differences in the underlying held-out sample rather than any change in the classification problem itself; nevertheless, both evaluation schemes converge on the same conclusion, a substantially richer 42-dimensional feature representation and L1-based feature selection each provide, at best, a marginal advantage over the baseline.

The two models also diverge in how they fail. Model 1's recall for "Like" trials (0.40) is only slightly lower than its recall for "Dislike" trials, whereas Model 2's L1 regularization drives recall for "Like" down further, to 0.14, while precision for the same class rises (0.65 $\pm$ 0.23). This pattern suggests that shrinking the feature set toward a sparser, more conservative representation makes the model even more reluctant to predict the minority "Like" class, trading a small gain in overall accuracy for a near-complete inability to detect participants' positive product evaluations.

Overfitting analysis across both models converges on the same conclusion. For Model 1, training and test accuracy were nearly identical (57.0% vs. 56.6%). Model 2 showed an equally small train-

test gap (58.0% $\pm$ 2.3% vs. 57.8% $\pm$ 2.6%, Table 4) and stable performance across all five outer folds (range: 53.6%–61.2%, Table 3). In both cases, the gap between training and held-out performance is far smaller than the gap between either model and a meaningfully useful classifier. This indicates that the models' limited accuracy is not an artifact of memorizing subject-specific training patterns; rather, it reflects a genuine ceiling on the amount of like/dislike information recoverable from trial-level EEG band-power features once evaluation is restricted to unseen subjects.

Taken together, these results reinforce the conclusion from Project I: individual differences in EEG activity continue to dominate over condition-level (like/dislike) differences, even when the feature space is expanded from a single FAA metric to a full 42-dimensional band-power representation and combined with L1 regularization and subject-grouped cross-validation. Adding more features and more principled regularization did not resolve the underlying signal-to-noise problem; it only marginally shifted where the models sit relative to baseline. This suggests that future modeling efforts might benefit less from further feature engineering within this framework and more from approaches that explicitly model or remove individual-level variance, such as subject-level standardization, mixed-effects modeling, or within-subject prediction.

5 Discussion and Conclusion

The logistic regression model achieved a higher accuracy than the majority-class baseline, indicating that EEG-derived spectral features contain predictive information related to consumer preference. Although the improvement was modest, the results suggest that neural activity distributed across multiple electrodes and frequency bands provides more useful information than frontal alpha asymmetry alone, which was not statistically significant in Project I. This finding supports the shift from statistical inference to machine learning for modeling complex EEG patterns associated with consumer decision-making. However, the model's performance remains limited, reflecting the inherent variability and noise of EEG signals as well as the restricted capacity of a linear classifier to capture complex neural relationships. In addition, the relatively small dataset and subject-to-subject variability may have constrained generalization to unseen individuals. Overall, the findings provide preliminary evidence that conventional EEG spectral features can support consumer preference prediction, but more advanced feature representations, larger datasets, and nonlinear machine learning models may be required to achieve practically useful prediction performance.

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